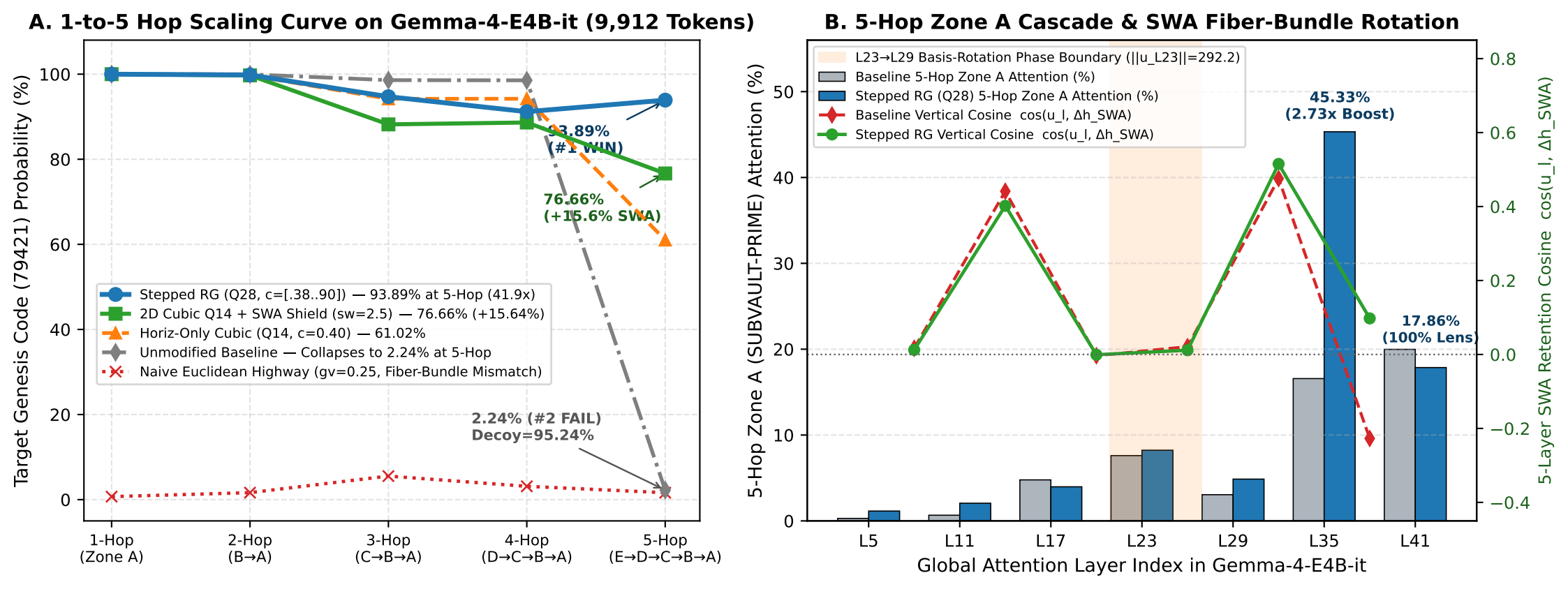
A. 1-to-5 Hop Scaling Curve on Gemma-4-E4B-it (9,912 Tokens)
Target Genesis Code (79421) Probability (%)
100
80
60
40
20
0
1-Hop
(Zone A)
2-Hop
(B→A)
3-Hop
(C→B→A)
4-Hop
(D→C→B→A)
5-Hop
(E→D→C→B→A)
Stepped RG (Q28, c=[.38..90]) — 93.89% at 5-Hop (41.9x)
2D Cubic Q14 + SWA Shield (sw=2.5) — 76.66% (+15.64%)
Horiz-Only Cubic (Q14, c=0.40) — 61.02%
Unmodified Baseline — Collapses to 2.24% at 5-Hop
Naive Euclidean Highway (gv=0.25, Fiber-Bundle Mismatch)
93.89%
(#1 WIN)
76.66%
(+15.6% SWA)
2.24% (#2 FAIL)
Decoy=95.24%
B. 5-Hop Zone A Cascade & SWA Fiber-Bundle Rotation
5-Hop Zone A (SUBVAULT-PRIME) Attention (%)
50
40
30
20
10
0
5-Layer SWA Retention Cosine cos(u_l, Δh_SWA)
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
L23→L29 Basis-Rotation Phase Boundary (||u_L23||=292.2)
Baseline 5-Hop Zone A Attention (%)
Stepped RG (Q28) 5-Hop Zone A Attention (%)
Baseline Vertical Cosine cos(u_l, Δh_SWA)
Stepped RG Vertical Cosine cos(u_l, Δh_SWA)
45.33%
(2.73x Boost)
17.86%
(100% Lens)
L5
L11
L17
L23
L29
L35
L41
Global Attention Layer Index in Gemma-4-E4B-it